

Lab 10.2.7 Report: Using Wireshark to Examine a UDP DNS Capture

Student Name: omar mohammad hamed

Part 1: Record VM's IP Configuration Information

Network Configuration Table:

Description	Settings
IP address	192.168.1.24
MAC address	08:00:27:7e:d0:17
Default gateway IP address	192.168.1.1
DNS server IP address	8.8.4.4

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Part 2: Use Wireshark to Capture DNS Queries and Responses

Successfully captured DNS query and response packets by:

1. Starting Wireshark on interface enp0s3
2. Navigating to www.google.com in web browser
3. Stopping capture after page loaded

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Part 3: Analyze Captured DNS or UDP Packets

Step 2: DNS Query Packet Analysis

Question 1: Is the source MAC address the same as recorded in Part 1?

Answer:

Yes, the source MAC address (08:00:27:7e:d0:17) matches the MAC address recorded in Part 1. This confirms that the DNS query originated from my virtual machine. The destination MAC address is the default gateway's MAC address because all packets leaving the local network must pass through the gateway first before reaching the DNS server on the internet.

Source and Destination Table:

Device	IP Address	MAC Address
Source Workstation	192.168.1.24	08:00:27:7e:d0:17
Destination	8.8.4.4	98:08:6a:f2:b3:f0

Wireshark Results Table:

Description	Wireshark Results
Frame size	74 bytes
Source MAC address	08:00:27:82:75:df
Destination MAC address	98:08:6a:f2:b3:f0
Source IP address	192.168.1.24
Destination IP address	8.8.4.4
Source port	4356
Destination port	53

Question 2: Is the source IP same as recorded in Part 1?

Answer:

Yes, the source IP address (192.168.1.24) is the same as the IP address I recorded in Part 1. This confirms that my virtual machine initiated the DNS query to resolve the domain name www.google.com to its IP address.

Question 3: Is destination IP same as default gateway?

Answer:

No, the destination IP address (8.8.4.4) is not the same as the default gateway IP address (192.168.1.24) recorded in Part 1. The destination IP address is the Google Public DNS server. However, the destination MAC address belongs to the default gateway because the packet must pass through the gateway to reach the DNS server on the internet. This demonstrates the difference between Layer 3 (IP) routing and Layer 2 (MAC) switching.

Step 3: DNS Response Packet Analysis

Question 4: What device is source/destination MAC in DNS response?

Answer: In the DNS response packet:

- Source MAC address: Default Gateway (98:08:6a:f2:b3:f0)
- Destination MAC address: My VM (08:00:27:7e:d0:17)

The roles have reversed compared to the DNS query. The response travels from the internet through the default gateway back to my virtual machine.

The gateway acts as an intermediary, receiving the response from the DNS server and forwarding it to my VM on the local network.

Question 5: What is the destination IP address?

The destination IP address is **192.168.1.24**, which is my virtual machine's IP address. The DNS server is sending the response back to the device that requested the domain name resolution.

Question 6: What is the source IP address?

Answer: The source IP address is **8.8.4.4**, which is the Google Public DNS server. This is the same DNS server that my VM queried in the previous step.

Question 7: What happened to source/destination roles?

Answer:

The roles of source and destination have completely reversed:

DNS Query:

- Source IP: My VM (192.168.1.24)
- Destination IP: DNS Server (8.8.4.4)
- Source MAC: My VM (08:00:27:7e:d0:17)
- Destination MAC: 98:08:6a:f2:b3:f0

DNS Response:

- Source IP: DNS Server (8.8.4.4)
- Destination IP: My VM (192.168.1.24)
- Source MAC: 98:08:6a:f2:b3:f0
- Destination MAC: My VM (08:00:27:7e:d0:17)

This reversal is normal in client-server communication. The response

follows the reverse path of the request, ensuring proper bidirectional communication.

Reflection Question

What are the benefits of using UDP instead of TCP for DNS?

Answer:

Using UDP instead of TCP for DNS provides several significant benefits:

1. Speed and Efficiency:

- UDP is connectionless and does not require a three-way handshake like TCP, making DNS queries much faster
- DNS queries and responses are typically very small (often less than 512 bytes), so the overhead of establishing a TCP connection would waste time and resources

2. Lower Overhead:

- UDP headers are only 8 bytes compared to TCP's 20-byte minimum header
- No connection state needs to be maintained on either client or server
- This allows DNS servers to handle many more simultaneous queries

3. Simplicity:

- UDP's stateless nature means DNS servers don't need to track connection states for thousands of clients

- If a DNS query is lost, the client simply sends another query
- This simplicity makes DNS infrastructure more scalable

4. Reduced Latency:

- For a TCP DNS query: SYN, SYN-ACK, ACK, Query, Response, FIN, ACK
(7 packets minimum)
- For a UDP DNS query: Query, Response (2 packets only)
- This dramatic reduction in packet count significantly reduces latency

5. Better Resource Utilization:

- DNS servers can handle more queries per second with UDP
- Less memory required as no connection state is stored
- Lower CPU usage without TCP's complex state management

6. Sufficient Reliability:

- DNS queries are idempotent (can be repeated safely)
- If no response is received, the application can simply retry
- For large responses (>512 bytes), DNS can fall back to TCP if needed

The only drawback is lack of reliability guarantees, but for DNS this is acceptable because:

- Queries are small and quick to resend
- DNS data is often cached, reducing the need for repeated queries
- The performance benefits far outweigh the occasional lost packet

In summary, UDP is perfectly suited for DNS because it prioritizes speed

and efficiency over reliability, which is exactly what DNS needs for optimal performance.

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Conclusion:

This lab provided hands-on experience with network protocol analysis using Wireshark. I learned how DNS queries work at the packet level, understanding the role of UDP as a transport protocol. By examining both query and response packets, I gained insight into how source and destination addresses and ports reverse in client-server communication. The lab demonstrated why UDP is preferred over TCP for DNS - its speed, efficiency, and low overhead make it ideal for small, frequent queries.

Key takeaways:

- DNS uses UDP port 53 for queries and responses
- MAC addresses change at each hop (Layer 2) while IP addresses remain constant (Layer 3)
- UDP's simplicity makes it perfect for DNS traffic
- Wireshark is a powerful tool for understanding network protocols

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